

Data-Driven Health Monitoring and Fault Detection for the RS-25 Staged Combustion Liquid Rocket Engine in EcosimPro/ESPSS framework

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Francesco Mattia⁽¹⁾, Marco Fabiani⁽¹⁾, Matteo Fiore⁽¹⁾, Francesco Nasuti⁽¹⁾

⁽¹⁾ Sapienza University of Rome, Rome, Italy

Email: mattia.1889824@studenti.uniroma1.it, m.fabiani@uniroma1.it, matteo.fiore@uniroma1.it, francesco.nasuti@uniroma1.it

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ABSTRACT:

This paper presents a health monitoring system for open-set fault recognition applied to the RS-25 liquid rocket engine. A digital twin developed in the EcosimPro/ESPSS framework is used to simulate both nominal and faulty operation, covering 24 failure modes of varying intensity. The system extends a standard closed-set classifier with a distance-based rejection mechanism, allowing unknown faults — not seen during training — to be flagged rather than misclassified. Among the evaluated classifiers, K-Nearest Neighbors proves the most robust, minimizing open space risk while maintaining high accuracy on known failure patterns. A multi-objective optimization is then performed to identify the optimal sensor subset: results show that 15 sensors suffice to correctly classify over 94% of both known and unknown failure modes, with excessive dimensionality degrading performance. The proposed framework serves both as a sensor placement tool during engine design and as a reliable real-time diagnostic system.

1. Introduction

Modern liquid rocket engines (LREs) serve as the critical power units for launch vehicles, requiring extreme reliability due to the harsh thermal and mechanical environments in which they operate [1]. To achieve economically sustainable space exploration, current industry trends focus on the development of reusable launch vehicles, which demand engines capable of wide-range throttling and repeated ignition [2, 3]. These operational complexities, combined with the need to minimize time-consuming manual inspections between flights, necessitate the implementation of advanced health monitoring systems. Such systems must be capable of real-time fault detection and diagnosis to identify malfunctions early, thereby preventing catastrophic mission failures and protecting expensive flight hardware [4].

The field of LRE health monitoring has evolved through several distinct stages since the Space Shuttle era. Initial efforts relied on red-line cut-off systems and the System for Anomaly and Failure Detection, which utilized statistical thresholds

for steady-state monitoring [5]. Subsequently, research shifted toward analytical model-based methods, leveraging energy balance equations and non-linear Kalman filters to provide physical insights into engine dynamics during both steady-state and transient phases, such as startup and shutdown [6, 7]. More recently, data-driven approaches using machine learning—including Long Short-Term Memory networks, Convolutional Neural Networks, and Support Vector Machines—have demonstrated superior performance in extracting features from multi-sensory data to identify complex failure modes [8–13]. Furthermore, researchers have explored Optimal Sensor Placement strategies to maximize the diagnosability of the system while minimizing the weight and cost associated with redundant instrumentation [10, 14].

Despite these advancements, a significant limitation remains: the vast majority of existing LRE fault detection literature operates under a closed-set assumption. In a closed-set scenario, the classifier is trained only on a predefined set of known failure classes, assuming that all possible anomalies encountered during testing will belong to those specific categories. This is unrealistic for complex propulsion systems, as it is nearly impossible to simulate or test every possible failure mode and intensity beforehand [11, 15–17]. When a closed-set system encounters an unknown or novel fault, it inevitably misclassifies it as one of the known types, potentially leading to incorrect emergency responses or hazardous maintenance decisions [18, 19].

This work addresses this gap by focusing on open-set fault detection, developing a health monitoring system capable of distinguishing between known failure modes and unknown anomalies [9, 19, 20]. The system is designed to provide a specific label if a fault matches a known signature or to reject the sample as "unknown" if it deviates from established patterns. To maximize performance, we incorporate sensor selection optimization tailored for open-set diagnosability. The proposed methodology is validated using the Space Shuttle Main Engine (SSME/RS-25) as a test case. The engine is reconstructed via a Digital Twin developed in the Ecosimpro/ESPSS simulation environment [21], allowing for the generation of diverse datasets comprising failures of varying types and intensities to rigorously evaluate the robustness of the open-set classifier.

2. Test Case

This section details the simulation strategy and the data generation process required to train and evaluate the diagnostic framework. First, the architecture of the target propulsion system and the corresponding simulation environment are presented. A baseline of nominal operations is then established through a Monte Carlo approach to account for inherent physical uncertainties. Subsequently, a dataset of anomalous conditions is generated by systematically injecting specific failure modes across the subsystems. Finally, the suite of observable telemetry available to the diagnostic algorithms is introduced.

2.1. SSME Architecture and Simulation Environment

The system selected as the primary test case for the proposed diagnostic framework is the Space Shuttle Main Engine (SSME) (Fig. 1). As illustrated in the schematic in Fig. 1, the SSME is a liquid-propellant rocket engine that operates on a closed-loop staged combustion cycle, utilizing Liquid Hydrogen (LH2) as fuel and Liquid Oxygen (LOX) as oxidizer.

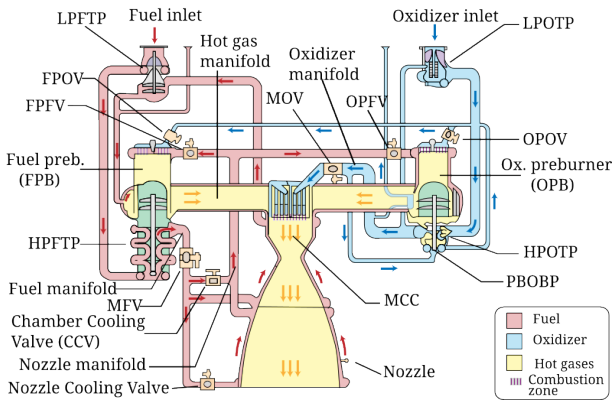


Figure 1: SSME schematic.

Due to the engine's inherent complexity and tight interconnections, a local anomaly can rapidly propagate throughout the entire system. To capture these dynamic responses and perform a comprehensive system-level analysis, a reduced-order model of the engine was developed using EcosimPro. EcosimPro is a versatile simulation environment widely used in Europe for the modeling and analysis of complex physical systems. Building on this platform, the ESPSS (European Space Propulsion System Simulation) toolkit has been developed by ESA to support the design, analysis, and performance evaluation of space propulsion systems [21]. This environment allows users to create modular, reduced-order components that can reproduce essential physical phenomena while maintaining computational efficiency. This capability is crucial for the proposed diagnostic framework, as it enables the

execution of the extensive simulation campaigns required to generate comprehensive datasets under both nominal and fault conditions, providing a physically sound yet computationally viable foundation for the data-driven models.

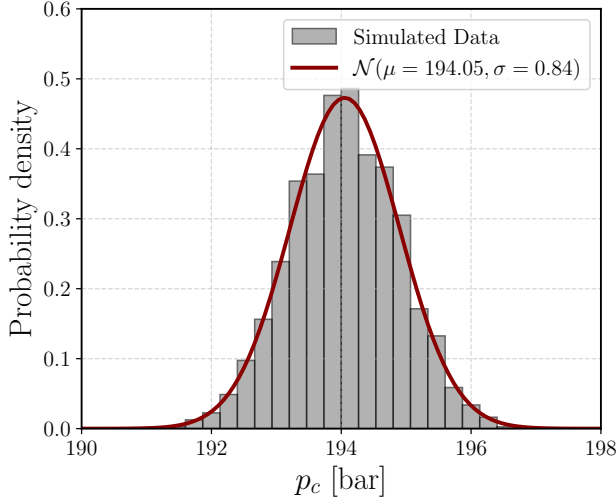
2.2. Nominal State Definition

In real-world operations, a “nominal” state is never a single deterministic operating point. Manufacturing tolerances, material degradation, ambient condition variations, and inherent sensor noise create a natural envelope of acceptable behaviors. Evaluating a diagnostic framework on a single ideal simulation would lead to a highly brittle baseline, prone to generating false alarms during standard operational variations.

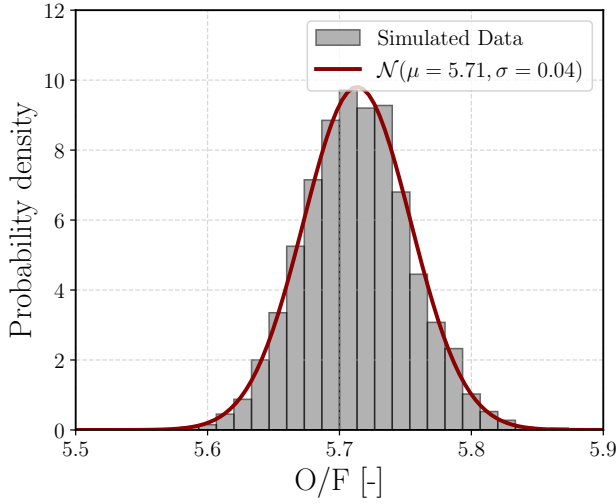
To address this, a baseline of the nominal state was generated through a Monte Carlo approach. Inherent uncertainties were injected into key physical parameters within the EcosimPro model. Specifically, statistical variations were applied to the geometric dimensions (e.g. cross-sectional flow areas), surface roughness of the internal channels and fluidic components, and temperatures of both propellants in the tanks. A total of $N_{\text{nom}} = 3000$ independent simulations were performed. In each run, these uncertain input variables were randomly sampled according to their prescribed statistical distributions. Notably, the simulation campaign focused exclusively on the steady-state nominal operating point, intentionally excluding the non-linear dynamics of the start-up and shut-down transients.

The propagation of these input uncertainties results in a natural variance across all measured thermodynamic outputs. As a representative example, Fig. 2 illustrates the probability density functions of key global performance indicators, such as the main combustion chamber (MCC) pressure (p_c) and the MCC mixture ratio (O/F). For each monitored quantity, the resulting realizations were used to construct normalized histograms and extract the corresponding sample mean (μ) and standard deviation (σ). A Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ was fitted to the samples, providing a direct visual check of the approximate normality of the propagated uncertainties.

Across the generated dataset, all variables of interest exhibit bell-shaped distributions with limited spread and an excellent match with the corresponding Gaussian fits. The tight clustering of critical parameters guarantees stable combustion conditions and robust overall performance. Consequently, these 3000 realizations establish a comprehensive “Nominal Cloud” representing the healthy engine operations. This multidimensional baseline envelope is directly visualized in the bivariate scatter plots of Fig. 3, which show the nominal dispersion and the physiological correlations between the coupled subsystems. Ultimately, this cloud provides the reference against which the subsequent anomaly detection algorithms will be evaluated.



(a) MCC pressure.



(b) MCC mixture ratio.

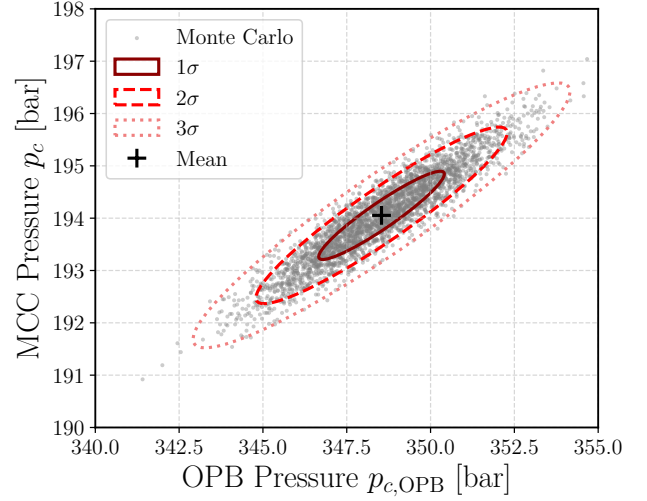
Figure 2: Probability density functions of key performance indicators obtained from the Monte Carlo simulations of the nominal state. The histograms representing the simulated dataset are overlaid with their corresponding Gaussian fits.

2.3. Fault Scenarios

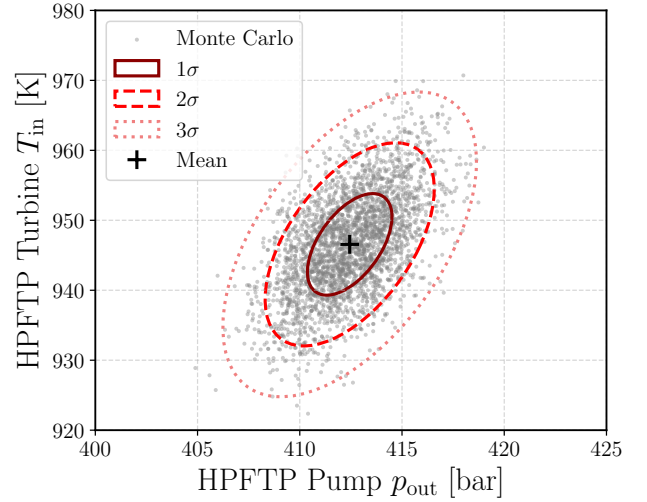
Once the robust nominal baseline was established, the subsequent phase involved generating a comprehensive anomaly dataset.

In the EcosimPro environment, each failure mode was introduced at the schematic level by perturbing a specific physical quantity. Crucially, to ensure a robust evaluation across different degradation severities, each failure was not simulated as a single discrete event, but rather tested across a predefined range of variable intensities. The implementation followed a parametric formulation where a failure-intensity parameter was systematically swept between its nominal baseline and a maximum degradation limit.

A total of 24 distinct failure scenarios were modeled to provide a structured and comprehensive



(a) System-level coupling: Main Chamber vs. Oxidizer Pre-burner pressures.



(b) HPFTP coupling: turbine inlet temperature vs pump discharge pressure.

Figure 3: Bivariate scatter plots illustrating the nominal operating envelope. The covariance ellipses (1σ , 2σ , and 3σ) are also reported.

exploration of the propulsion system's off-nominal conditions. As detailed in the dedicated summary reported in Table 1, these anomalies cover a wide spectrum of physical degradation mechanisms. Valve actuation anomalies and leakages were simulated for the primary control components, encompassing the abnormal opening and internal leakage of the FPOV, MFV, MOV, and OPOV. Performance degradations within the turbomachinery were modeled not only through efficiency reductions across the low-pressure and high-pressure stages for both pumps and turbines, but also by introducing geometric restrictions via the reduction of turbine inlet effective areas. Furthermore, the simulation campaign incorporated severe system-level faults, such as manifold leakages, MCC cooling circuit breaches and occlusions.

Following the overview of the simulated anoma-

Table 1: Summary of the 24 simulated fault scenarios, categorized by subsystem, component, failure mode and intensity.

Subsystem	Component	Failure Mode	Intensity Range
Control Valves	MFV (main fuel valve)	Abnormal Opening Leakage	1 – 100% 0 – 47%
	MOV (main oxidizer valve)	Abnormal Opening Leakage	1 – 100% 0 – 98%
	FPOV (fuel pre-burner oxidizer valve)	Abnormal Opening Leakage	1 – 100% 0 – 54%
	OPOV (oxidizer pre-burner oxidizer valve)	Abnormal Opening Leakage	1.3 – 100% 0 – 48%
Fuel Turbopumps	LPFTP (low-pressure fuel turbopump)	Turbine Efficiency Pump Efficiency Turbine Inlet Area Red.	0.12 – 0.52 0.25 – 0.65 10 – 90%
	HPFTP (high-pressure fuel turbopump)	Turbine Efficiency Pump Efficiency Turbine Inlet Area Red.	0.38 – 0.78 0.33 – 0.73 15 – 55%
Oxidizer Turbopumps	LPOTP (low-pressure oxidizer turbopump)	Turbine Efficiency Pump Efficiency Turbine Inlet Area Red.	0.40 – 0.80 0.23 – 0.63 10 – 90%
	HPOTP (high-pressure oxidizer turbopump)	Turbine Efficiency Pump Efficiency Turbine Inlet Area Red.	0.38 – 0.78 0.21 – 0.61 10 – 65%
	PBOBP (pre-burner oxidizer boost pump)	Pump Efficiency	0.40 – 0.80
Others	Hot Gas Manifold	Leakage	0 – 92%
	Oxidizer Manifold	Leakage	0 – 70%
	MCC Cooling Jacket	Occlusion Leakage	0 – 58% 0 – 70%

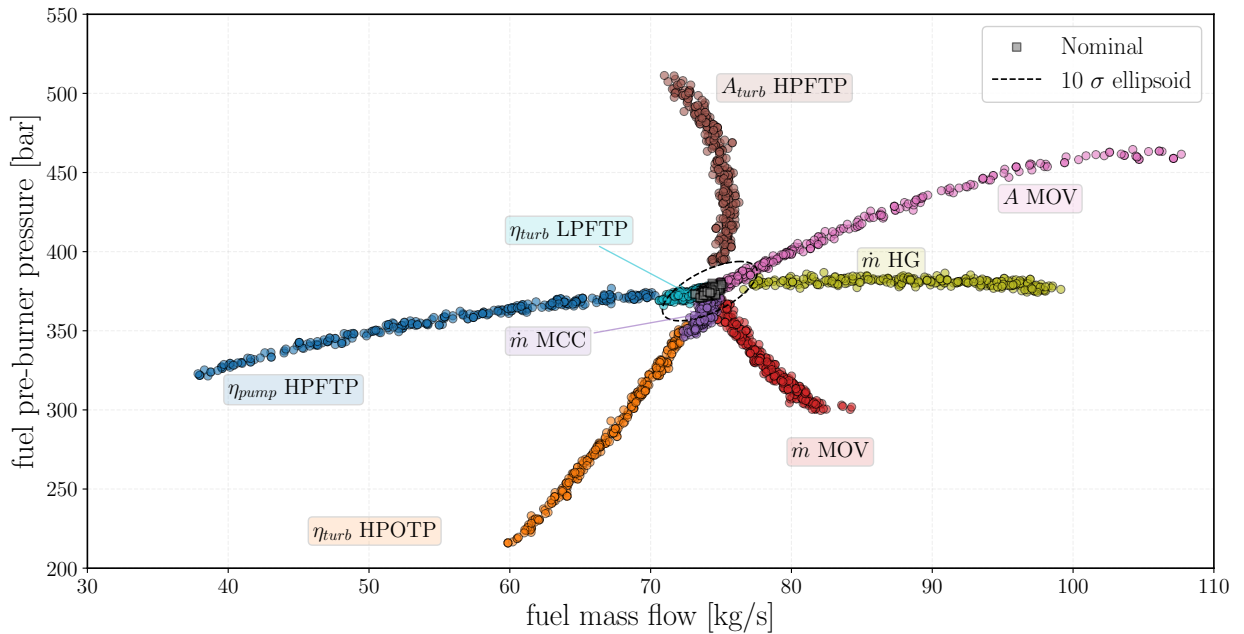


Figure 4: Representative fault signatures depicting the engine's off-nominal behavior. A represents a reduction in cross-section, $\dot{m}$ a leakage, and η a decrease in efficiency.

lies detailed in Table 1, representative examples of the system's dynamic response are provided to illustrate the engine's off-nominal behavior in Fig. 4. Two sensors are selected, namely the fuel mass flow rate and the pressure of the fuel pre-burner. It can be seen how all failures start near the nominal operating point (low intensity failure) and as the intensity increase induce a significant change in the operating point.

2.4. Telemetry Selection and Observability

The raw simulation output of EcosimPro was pruned to retain only observable telemetry. Parameters representing performance (e.g. thrust, specific impulse, mixture ratio, turbine power output) and redundant quantities (e.g. rotational speeds of pumps and turbines mounted on the same shaft) were excluded. This process resulted in a candidate set of 84 sensors, consisting of pressures, temperatures, mass flow rates and rotational speeds. This ensures the diagnostic framework relies exclusively on features detectable by standard flight instrumentation.

3. Methodology

This section describes the diagnostic pipeline developed to address the open-set fault detection problem. The raw simulation data are first standardized and enriched through a kernel density estimation transformation, as detailed in Section 3.1. Three candidate classifiers are then trained and compared in an open-set regime, each equipped with a threshold-based rejection mechanism to flag anomalies that deviate from any known failure signature (Sections 3.2-3.3). Finally, Section 3.4 introduces the evaluation metrics used to jointly assess classification accuracy on known faults and rejection capability on unknown ones.

3.1. Data Pre-Processing

The raw measurements of sensor j (x) are pre-processed before classification. First of all, a zero-mean unit-variance scaler is fitted on the nominal data

$$\tilde{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{N_{\text{nom}}} \sum_{i=1}^{N_{\text{nom}}} x_{i,j} \quad (1)$$

$$\sigma_j = \sqrt{\frac{1}{N_{\text{nom}} - 1} \sum_{i=1}^{N_{\text{nom}}} (x_{i,j} - \mu_j)^2}$$

The same (μ_j, σ_j) statistics are applied without re-fitting to the test matrix, comprising both known and unknown failures. To enrich the feature representation with class-conditional density information, a KDE transformation is applied after normalization. For each known class $c \in \mathcal{K}$ and each sensor channel $s \in \{1, \dots, d\}$, a one-dimensional Gaussian KDE

is fitted on the fit-set samples of that class to obtain a probability density function $\hat{p}$:

$$\hat{p}(x | c, s) = \frac{1}{N_{c,s} h} \sum_{i \in \mathcal{I}_{c,s}} K\left(\frac{\tilde{x} - \tilde{x}_{i,s}}{h}\right) \quad (2)$$

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$$

where h is the bandwidth hyperparameter (set to 0.5) and $\mathcal{I}_{c,s}$ is the index set of fit-set samples belonging to class c , of size $N_{c,s}$. Each normalized sample $\tilde{\mathbf{x}} \in \mathbb{R}^d$ is therefore transformed into a $d \times |\mathcal{K}|$ vector of per-class likelihoods. The resulting feature vector encodes how well each sensor reading is explained by each known failure distribution, providing a richer discriminative basis than the raw standardized readings.

3.2. Classifier Training

Three discriminative classifiers are evaluated in the open-set regime:

k -Nearest Neighbours (kNN)

A kNN index is built over the fit-set feature matrix. For a test sample $\mathbf{z}$, let $\{(\mathbf{z}^{(j)}, c^{(j)})\}_{j=1}^k$ be its $k = 5$ nearest neighbours. The predicted class is that of the single nearest neighbour, and the confidence score is defined as the negative mean Euclidean distance to the k neighbours:

$$s_{\text{kNN}}(\mathbf{z}) = -\frac{1}{k} \sum_{j=1}^k \|\mathbf{z} - \mathbf{z}^{(j)}\|. \quad (3)$$

Higher scores indicate greater proximity to the training manifold.

Support Vector Machine (SVM)

A multi-class SVM with radial basis function kernel is trained using the one-vs-rest decomposition. The confidence score is the maximum signed margin across all binary classifiers:

$$s_{\text{SVM}}(\mathbf{z}) = \max_{c \in \mathcal{K}} g_c(\mathbf{z}), \quad (4)$$

where $g_c(\mathbf{z})$ is the decision function of the binary classifier for class c .

Random Forest (RF)

An ensemble of 100 decision trees is trained. The class-posterior vector $\hat{\mathbf{p}}(\mathbf{z})$ is estimated by majority vote across trees, and the confidence score is its maximum component:

$$s_{\text{RF}}(\mathbf{z}) = \max_{c \in \mathcal{K}} \hat{p}(c | \mathbf{z}). \quad (5)$$

All three classifiers operate in the same normalized feature space, so their confidence scores are directly comparable once calibrated.

3.3. Threshold-Based Rejection (Open-Set Calibration)

All confidence scores defined in Section 3.2 share the same semantics: a higher value means the classifier is more certain that the sample belongs to a known class. A sample is labelled unknown whenever its score falls below a per-class rejection threshold. The threshold for class c is derived from the validation set:

$$\tau_c = \text{Percentile}_p(\{s(\mathbf{z}_i)\}_{i:y_i=c, i \in \mathcal{V}}), \quad (6)$$

where $\mathcal{V}$ denotes the validation partition and $p = 5$ (i.e., the 5th percentile). At inference time, the open-set prediction is

$$\hat{y}(\mathbf{z}) = \begin{cases} \hat{c}(\mathbf{z}) & \text{if } s(\mathbf{z}) \geq \tau_{\hat{c}(\mathbf{z})}, \\ \text{Unknown} & \text{otherwise,} \end{cases} \quad (7)$$

where $\hat{c}(\mathbf{z})$ is the closed-set predicted class. The percentile-based calibration is deliberately conservative: on average, p % of correctly classified validation samples would be rejected, establishing a controlled trade-off between false rejections and missed unknowns. A value of $p = 5$ was adopted.

3.4. Evaluation Metrics

Performance is assessed on two complementary metrics that capture distinct aspects of the open-set diagnostic problem.

Known-Known-Closed Macro-F1 (F1 KKC)

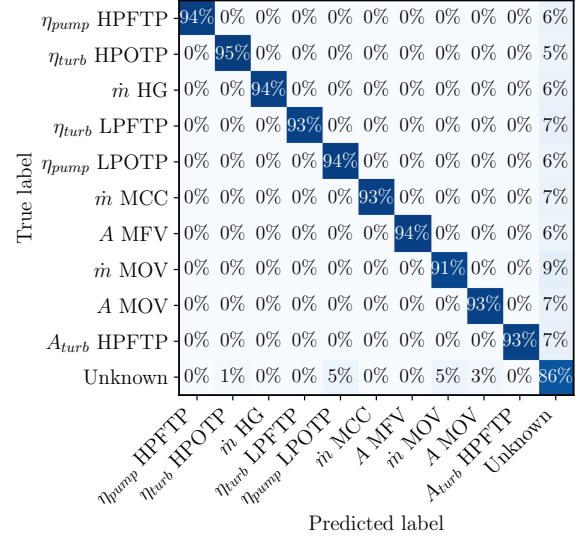
This metric evaluates how accurately the classifier identifies and distinguishes among the known failure modes, ignoring unknown test samples entirely. It is computed as the unweighted average of the per-class F1 scores across all known classes $\mathcal{K}$, so that rare failure modes contribute equally to the final score regardless of their frequency in the test set.

Unknown Recall

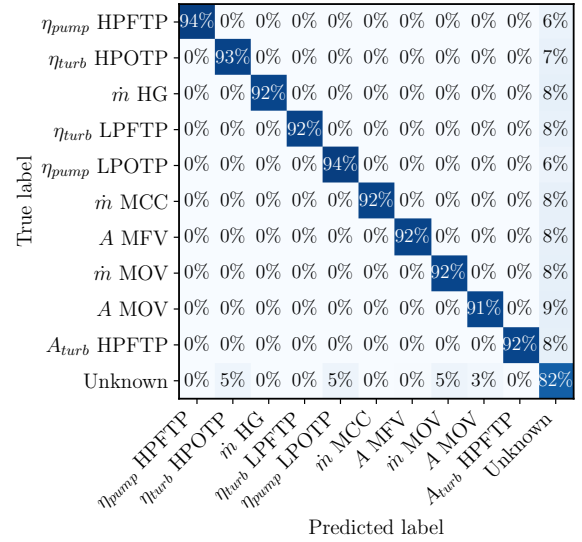
This metric evaluates the complementary capability of the system: its ability to recognize that a sample does not belong to any known failure class. Specifically, it measures the fraction of truly unknown samples — those belonging to classes never seen during training — that are correctly flagged as Unknown rather than misassigned to a known category.

4. Results

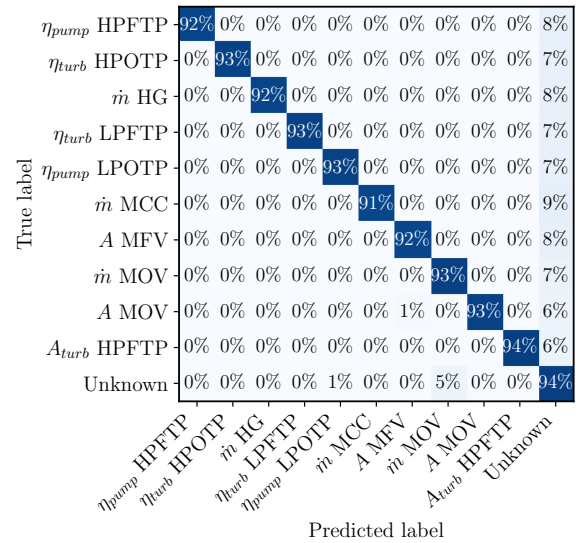
This section presents the results of the diagnostic pipeline developed to address the open-set fault detection problem. A comparison of different classifiers is performed in Sec. 4.1, and an in-depth assessment of the performance on unknown failure modes is reported in Sec. 4.2.



(a) Random Forest.



(b) Support Vector Machine.



(c) K-Nearest Neighbors.

Figure 5: Confusion matrices for the three candidate classifiers on a fixed set of known and unknown failure modes.

4.1. Classifier Selection

The selection of the optimal classification algorithm was conducted through two distinct analyses. First, the performance of three candidate models—K-Nearest Neighbors, Support Vector Machine, and Random Forest—was evaluated on a static set of failure modes to assess baseline diagnostic capabilities. Second, a parametric sensitivity study was performed to evaluate classifier robustness as the number of known failure classes increases, simulating the expansion of the system's knowledge base.

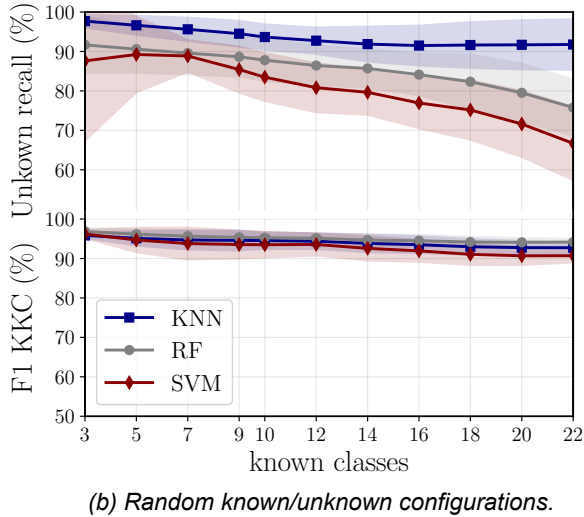
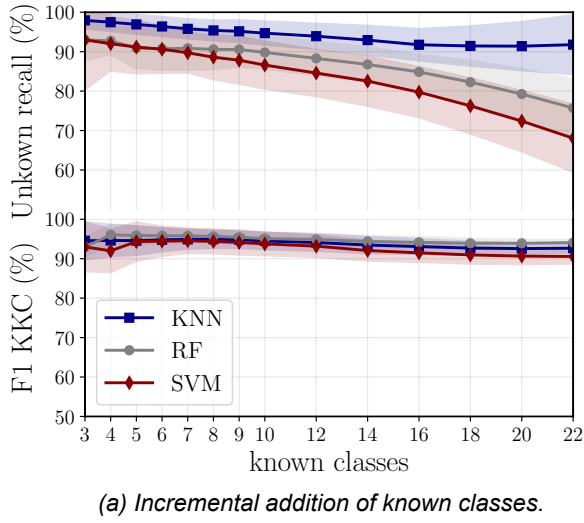


Figure 6: Classifier performance sensitivity to the number of known failure classes.

In the first analysis, a subset of 10 failure modes is arbitrarily selected as known classes (KKC), while the remaining modes are treated as unknown. The failures, shown in the labels of Fig. 5, have been selected to cover different subsystems, but not optimized to maximize performance. The classifiers are trained on the KKC and tested on the full dataset. To account for the stochastic nature of the training-test split, the process is repeated across multiple seeds.

The results, illustrated in the confusion matrices in Fig. 5, reveal divergent behaviors. For known failures, RF exhibits slightly higher accuracy due to its ability to define complex non-linear decision boundaries. However, KNN demonstrates superior performance in identifying unknown failures. This discrepancy is rooted in the *open space risk* [20]: while RF and SVM tend to over-partition the feature space into closed regions—effectively forcing outliers into known categories—KNN relies on local density and distance metrics. This local approach ensures that samples far from the training manifold are correctly rejected rather than misclassified.

The second analysis investigates how diagnostic performance scales with the complexity of the "known" set. This was addressed using two methodologies. In the first approach (Fig. 6a), a fixed pool of failures was used, and classes were added to the known set incrementally. This analysis was repeated 30 times to ensure statistical significance relative to the failure distribution. In the second approach (Fig. 6b), various random configurations of known/unknown ratios were tested to observe general trends.

As shown in Fig. 6, while the accuracy on KKC remains stable, the recall for unknown failures degrades as the number of known failures increases. This trend is expected: as the number of "labeled" classes grows, known regions occupy a larger portion of the feature space, increasing the likelihood that an unknown failure overlaps with a known decision boundary. Notably, KNN proved significantly more robust than SVM and RF, maintaining a higher rejection rate across all configurations. Consequently, KNN was selected as the core classifier for the subsequent stages of this study.

4.2. Unknown failure recognition

In this section, the capability of the KNN classifier to correctly reject unknown anomalies is investigated in detail. Following the setup of the first analysis in Section 4.1, the classifier is trained on a fixed pool of 10 known failure modes, while its performance is tested against 14 unknown failures. The objective is to identify which unknown modes are most frequently misclassified as known and to provide a physical justification for these occurrences.

The results show that the system achieves high robustness, with 11 out of 14 unknown failures exhibiting a recall higher than 99%. However, three specific failure modes show lower recall: leakage from the LOX Manifold (17%), efficiency decrease of the LPOTP turbine (90%), and efficiency decrease of the PBOBP pump (95%).

As illustrated in the feature space visualization (Fig. 7), the performance degradation for the latter two is primarily due to their low failure intensity. These samples lie within the 10σ threshold from the nominal operating point, a region where the phase space is extremely crowded. In this "near-nominal"

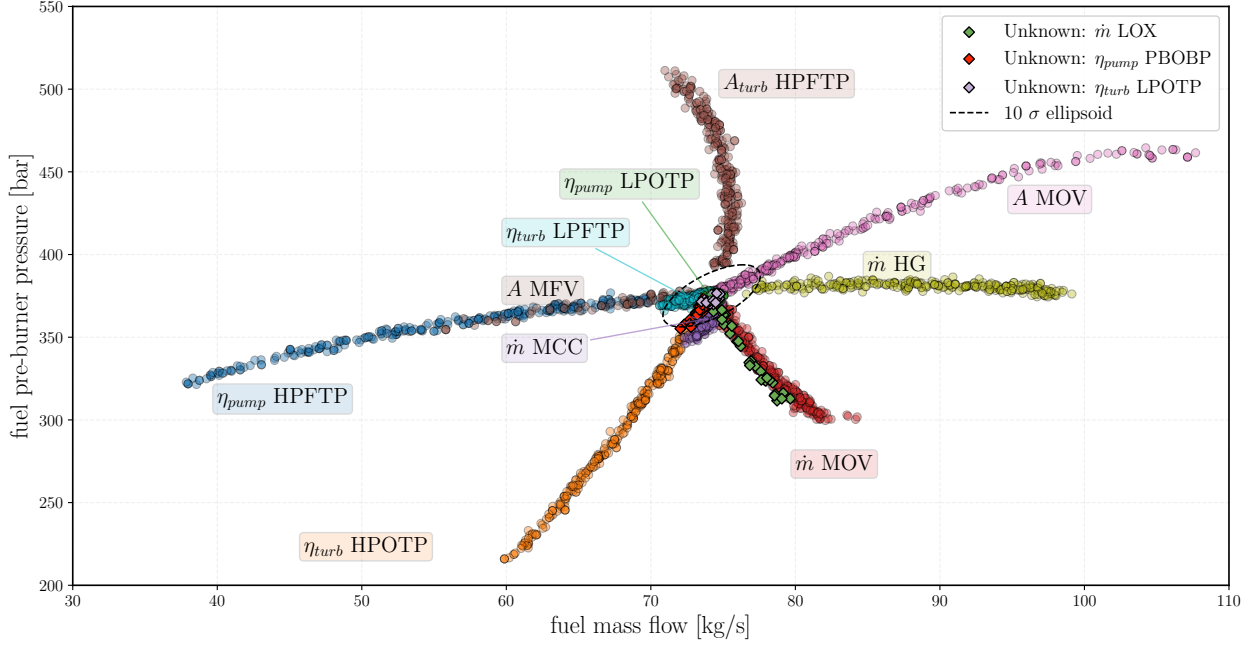


Figure 7: Scatter plot of known (dots) and unknown (diamonds) failures. Only unknown failures with recall lower than 99% are shown.

zone, the signatures of different failures converge, making it inherently difficult for any distance-based classifier to distinguish a subtle unknown anomaly from the manifold of known classes or healthy operation.

The case of the LOX manifold leakage presents a different challenge. Despite reaching higher intensities, it is consistently confused with the leakage from the main oxidizer valve (MOV). This behavior is physically consistent with the engine architecture: these two components are connected in series, and the absence of a flow sensor between them renders their effects on the downstream thermodynamic parameters nearly identical. From a data-driven perspective, these failures are virtually indistinguishable due to limited sensor observability.

In summary, this analysis demonstrates that open-set misclassifications in the HMS are driven by two main factors: (i) low fault intensity, which leads to overlapping signatures near the nominal state, and (ii) physical indistinguishability caused by the lack of dedicated sensors between components in series.

4.3. Optimal sensor placement

The final stage of the analysis investigates the impact of sensor suite configuration on diagnostic performance. Given the strict mass and telemetry constraints in space applications, identifying the minimal set of sensors that maximizes both known failure classification and unknown rejection is critical. To address this multi-objective optimization problem, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [22] was employed. This evolutionary algorithm allows for the identification of a Pareto front

for each fixed number of sensors (N_{sens}), representing the optimal trade-offs between the F1-score (on known classes) and the Recall (on unknown classes). The use of a Pareto-based approach provides flexibility to mission designers, allowing for the selection of operating points tailored to specific mission phases, such as prioritizing safety in crewed flights versus performance in uncrewed tests.

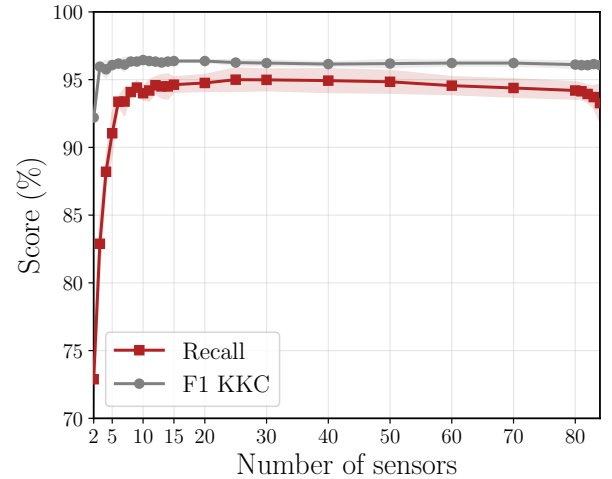
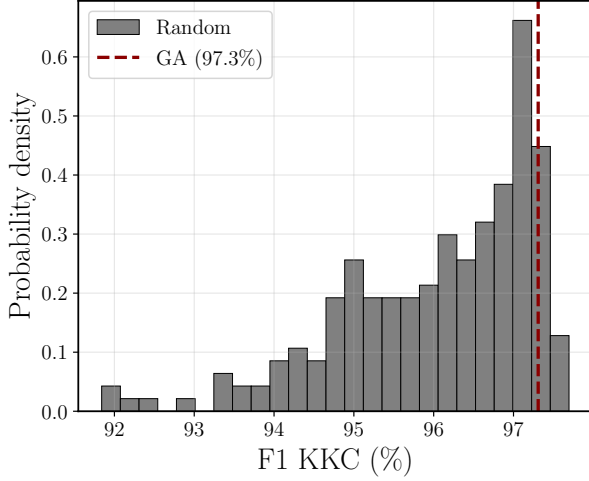
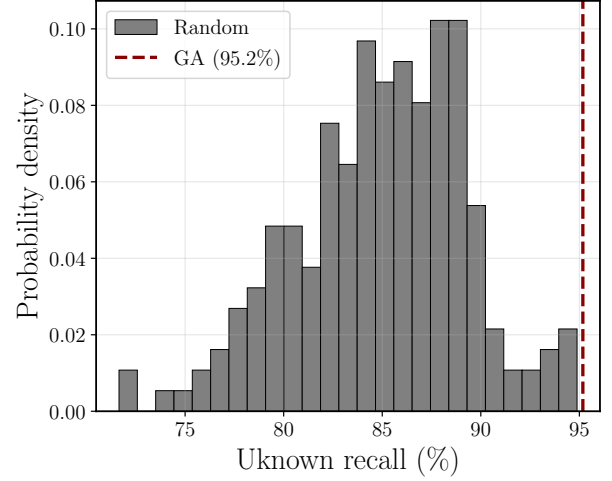


Figure 8: Diagnostic performance as a function of the number of sensors. The plot shows the Pareto knee points for F1-score (KKC) and Recall (Unknowns).

The evolution of the Pareto knee points as a function of the number of sensors is illustrated in Fig. 8. The results show that the F1-score for known failures reaches high stability ($\geq 97\%$) very rapidly, re-



(a) Known F1-score comparison.



(b) Unknown Recall comparison.

Figure 9: Comparison between the GA-optimized solution and 200 random sensor combinations for $N_{\text{sens}} = 20$.

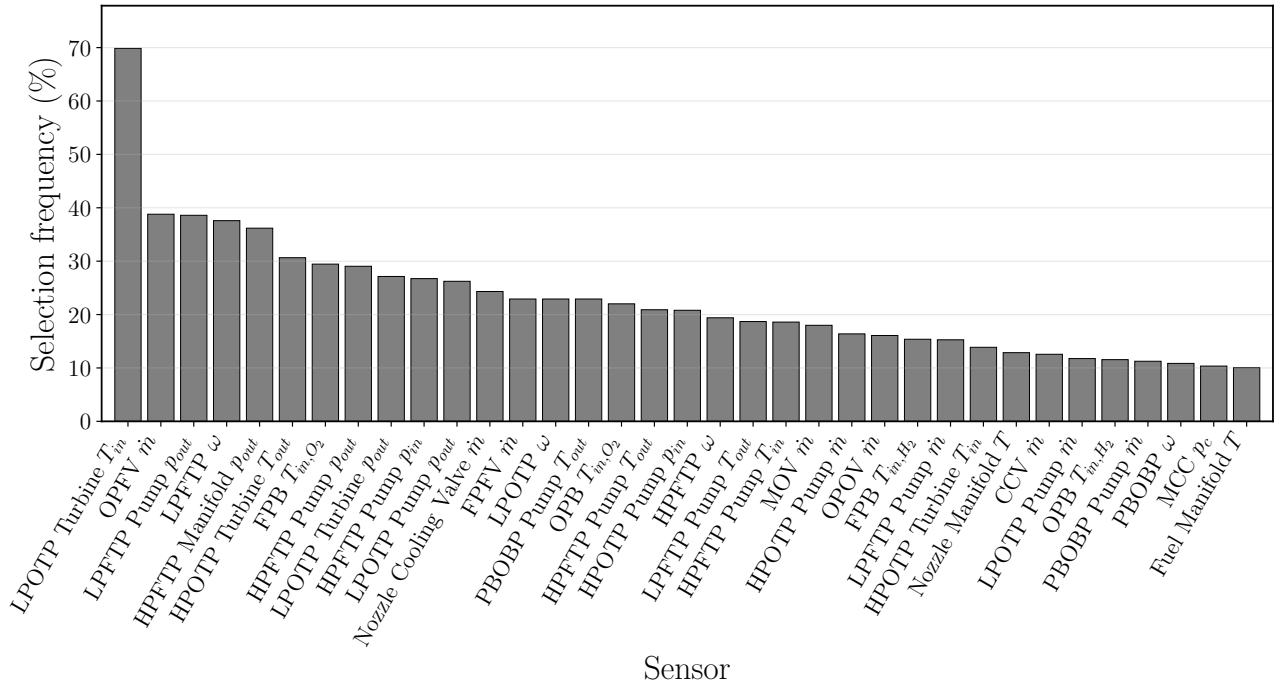


Figure 10: Selection frequency across all Pareto-optimal solutions for $N_{\text{sens}} < 20$. Only sensors with selection frequency $> 10\%$ are shown.

quiring only a few sensors. In contrast, the recognition of unknown failures (Recall) is more demanding, requiring approximately 10 to 15 sensors to plateau. Notably, a performance degradation is observed when the number of sensors approaches the full suite ($N_{\text{sens}} = 84$). This is attributed to the “curse of dimensionality” and the inclusion of structurally redundant or quasi-constant sensors (e.g., inlet temperatures with negligible variance), which introduce noise and distort the KNN distance metric, thereby reducing the classifier’s ability to define clear rejection boundaries.

To validate the effectiveness of the genetic optimization, Fig. 9 compares the GA-optimized so-

lution for $N_{\text{sens}} = 20$ against 200 random sensor combinations. The comparison highlights that while the F1-score on known classes is relatively insensitive to the specific sensor choice, the unknown Recall varies drastically across different combinations. This confirms that the ability to detect novel anomalies is highly dependent on the selection of specific discriminative features that characterize the engine’s global health state.

Finally, the selection frequency of sensors across all Pareto-optimal solutions is presented in Fig. 10. The ranking indicates that the GA consistently favors a specific subset of sensors regardless of the total number allowed. The most frequently selected sen-

sors coincide with those monitoring key thermodynamic nodes of the staged combustion cycle, while the "worst offenders"—not shown in the figure—are systematically discarded due to their lack of variance and redundant information, which provides no added value to the diagnostic manifold.

5. Conclusions

In this study, a data-driven diagnostic framework was developed to address the challenges of open set fault detection in liquid rocket engines. Leveraging the Ecosimpro/ESPSS framework, a comprehensive dataset of nominal and failure states were generated for the RS-25 staged combustion engine.

A comparative analysis revealed that while discriminative models like Random Forest excel in classifying known failures, the K-Nearest Neighbors algorithm is significantly more effective at identifying novel anomalies. This superiority is attributed to its distance-based decision boundaries, which effectively manage the open space risk by rejecting samples far from the training manifold. Detailed investigation into unknown failures highlighted that misclassifications are primarily driven by low-intensity faults near the nominal operating point or by physical indistinguishability due to sensor-limited observability, particularly in series-connected components.

Furthermore, the NSGA-II genetic algorithm was employed for optimal sensor placement, demonstrating that high diagnostic reliability does not require a full sensor suite. Peak performance in unknown rejection was achieved with approximately 15 sensors, after which the inclusion of redundant or low-variance sensors led to a "curse of dimensionality" effect that distorted the classifier's metric space. These findings provide a clear roadmap for the design of lean, robust health monitoring systems for future reusable launch vehicles, where minimizing telemetry and mass is as critical as ensuring mission safety.

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